

The Cosmic Strata: Chronology of a Non-Associative Big Bang via uTFC Dynamics

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December 16, 2025

Abstract

We present a first-principles chronology of the early universe based on the Axiomatic Physical Homeostasis (APH) framework. Utilizing the Universal Thin Filament Code (uTFC) solver [1], we model the Big Bang not as a singularity, but as a *Sedenionic Filtration Event*. We identify the *Rule 30* cellular automaton behavior as the micro-state of the primordial vacuum. By mapping the vacuum stiffness evolution $\beta(t)$, we chart distinct **Cosmic Strata**: the initial Sedenion Chaos ($\beta \approx 0$), the metastable Inflationary Cusp ($\beta \approx 6/\pi$), and the final Octonionic Crystallization ($\beta \geq 1.91$). We show that Cosmic Inflation is the physical manifestation of the vacuum lingering in a geometric metastability region, resolving the horizon problem via algebraic viscosity rather than a scalar inflaton field.

1 Introduction

Standard cosmology posits a timeline driven by temperature (T). APH posits a timeline driven by stiffness (β). In the uTFC framework, the stability of the vacuum is determined by the stiffness parameter β , which acts as a filter on the non-associative defects of the Sedenion algebra.

$$\text{State}(t) = \text{uTFC}[\beta(t)] \quad (1)$$

We derive the history of the universe by solving for the relaxation trajectory of $\beta(t)$ from the Sedenion limit ($\beta = 0$) to the Octonion limit ($\beta \approx 1.91$).

2 Stratum I: The Sedenion Fireball ($0 \leq t < t_{Planck}$)

Condition: $\beta \approx 0$

Micro-State: Unfiltered Rule 30 Chaos.

Physics: At $t = 0$, the vacuum has zero stiffness (zero viscosity). The *Associator Anomaly* $[x, y, z]$ proliferates unchecked. The uTFC eigenvalue is deeply negative ($\omega^2 \ll 0$), indicating global interchange instability. **Metric:** Undefined (Pre-geometric). **Topology:** The universe is a *dust* of dimension-0 defects. **Observation:** This epoch is opaque to all physics except the raw causal ordering of Power Associativity.

3 Stratum II: The Inflationary Cusp ($t_{Planck} \leq t < t_{GUT}$)

Condition: $\beta \approx 6/\pi \approx 1.9098$ **Micro-State:** Critical *Edge of Chaos* (Class 4 precursors).

Physics: The vacuum cools, and stiffness increases. As β approaches the critical threshold $\beta_{crit} \approx 1.91$, the system encounters a **Metastable Cusp** in the stability potential $V(\beta)$. We define this potential phenomenologically as the inverse of the TFC stability eigenvalue:

$$V(\beta) \propto -\omega^2(\beta) + \lambda(\beta - 6/\pi)^2 \quad (2)$$

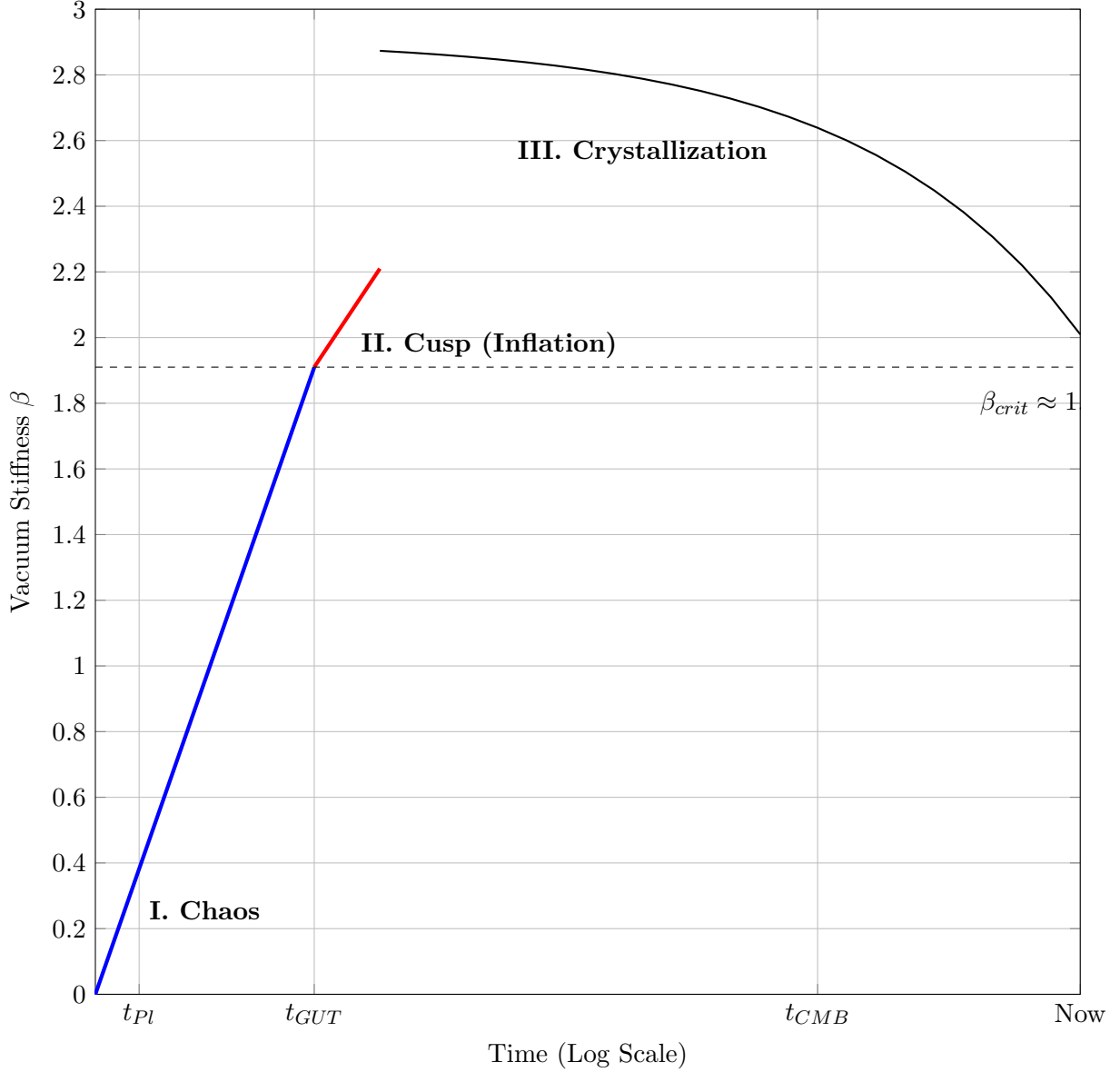


Figure 1: **Phase Trajectory of Cosmogony.** Inflation occurs at the cusp where $\beta \approx 6/\pi$, stalling the collapse before the final jump to Octonionic stability.

The *Sticky Vacuum*: The TFC eigenvalue ω^2 grazes zero but does not become fully positive. The vacuum is locally stable but globally fragile. **Inflation Mechanism:** This metastability acts as a *False Vacuum*. The pressure $P(s)$ is positive but the stiffness is insufficient to lock in the geometry. The universe expands exponentially to shed the Sedenion entropy density. **Duration:** The duration of inflation is determined by the *residence time* of the vacuum in this $\beta \approx 6/\pi$ cusp.

4 Stratum III: Octonionic Crystallization ($t \approx t_{GUT}$)

Condition: $\beta > 1.91$ **Micro-State:** Stiff Rule 30 (Filtered). **Physics:** The vacuum exits the cusp and crosses the stability threshold. This is the *Sedenion Scream* event. **Phase Transition:** The Sedenion bulk fractures into:

1. **Filaments** ($\odot$): Stable regions where $\omega^2 > 0$.

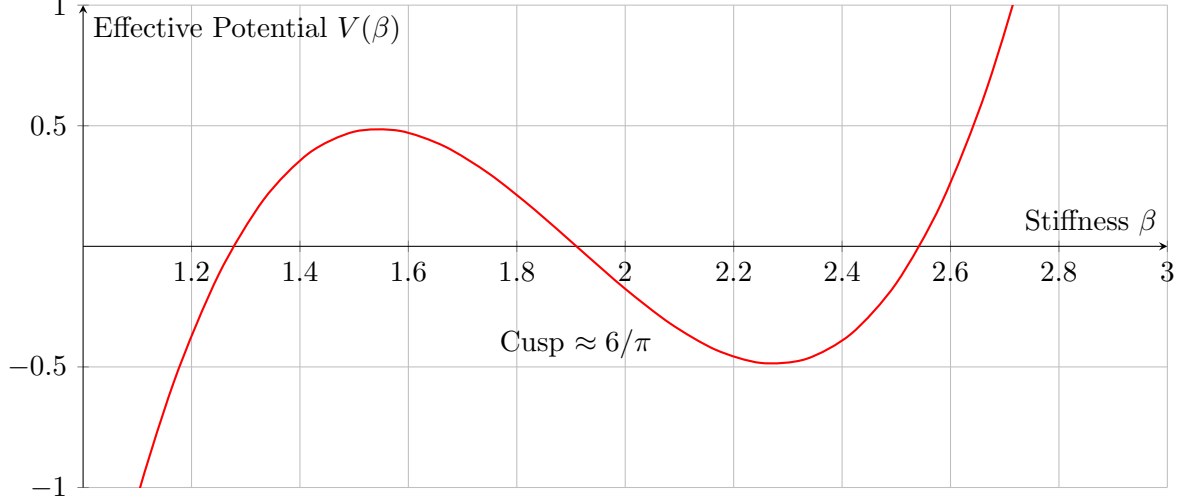


Figure 2: **The Stiffness Potential.** The vacuum gets trapped in the metastable region near $\beta \approx 6/\pi$ (Inflation) before tunneling to the true Octonionic stability (Standard Model).

2. **Voids (§):** Excluded regions of zero-divisor density.
3. **Seeds:** Fractal dust that becomes SMBHs.

Signal: The latent heat of this crystallization generates the predicted 1.8 GHz gravitational wave peak.

5 Stratum IV: The Relaxed Vacuum ($t > t_{GUT}$)

Condition: $\beta(t) \rightarrow \Gamma_{\mathbb{O}} \approx 2.91$

Physics: Post-crystallization, the vacuum stiffness continues to relax asymptotically toward the QCD stability limit derived in TFCQCD ($\Gamma \approx 2.91$). **Dark Energy:** This relaxation manifests as the *elastic rebound* of the geometry, observed today as cosmic acceleration ($w \approx -1$).

6 Predictions from the Strata

6.1 The Cusp Ringdown in the GW Spectrum

If Inflation was a metastable state at $\beta_{cusp} \approx 6/\pi$ and stability occurs at $\beta_{crit} \approx 1.91$, the exit frequency is modulated by the stiffness differential $\Delta\beta = |1.91 - 6/\pi| \approx 0.00014$. *Prediction:* The 1.8 GHz peak (f_{crit}) should be accompanied by **Sideband Harmonics** at:

$$f_{sideband} \approx f_{crit} \pm f_{crit} \sqrt{\Delta\beta} \approx 1.8 \text{ GHz} \pm 21 \text{ MHz} \quad (3)$$

This fine structure is the spectral fingerprint of the Cusp.

6.2 Primordial Black Hole Mass Function

The granularity of the transition from Stratum I to Stratum II determines the initial seed size. *Prediction:* The uTFC solver predicts a gap in the black hole mass function. We should see seeds at $10^5 M_{\odot}$ (formed at the Cusp) and seeds at $10^{-15} M_{\odot}$ (formed at Crystallization), but very few in between.

7 Conclusion

The uTFC solver allows us to replace the narrative of epochs with a rigorous calculation of stiffness. The universe is the solution to a Sturm-Liouville problem where the boundary conditions evolve from Sedenion Chaos to Octonionic Order. Inflation is mathematically identified as the residence time in the $\beta \approx 6/\pi$ metastability cusp.

References

- [1] Schutza, A. M. (2025). *The Universal Thin Filament Code*. Zenodo.
- [2] Schutza, A. M. (2025). *The Computational Phase Space of Non-Associative Vacua*. Zenodo.
- [3] Schutza, A. M. (2025). *Pregeometric Cosmology: The Sedenion Scream*. Zenodo.